ANSH AHUJA

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Education

Indian Institute of Information Technology Vadodara AUG 2024 – JUN 2028
BTech Computer Science CGPA: 7.5

Technical Skills

CS Fundamentals: Data Structures and Algorithms (700+ problems solved), Object-Oriented Programming (OOP), Operating Systems, DBMS
Languages: Java, JavaScript, TypeScript, Python, C/C++, SQL
AI / LLM: Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Agentic AI Workflows, Prompt Engineering, Embeddings, Tool/Function Calling, LLM APIs (OpenAI, Gemini), Hugging Face
Frameworks / Libraries: React.js, Next.js, Node.js, Express.js, TailwindCSS, Shadcn UI, Socket.IO
Databases / ORMs: MongoDB, MySQL, Vector Databases (Pinecone, ChromaDB, FAISS)
DevOps / Tools: Git/GitHub, Redis, Postman, Terraform

Experience

Mentora *Sep 2026 – Present*
Backend Developer Intern

- Built a **WebRTC video calling** service with Socket.IO signaling (SDP/ICE exchange) for low-latency peer-to-peer sessions.
- Developed a **real-time chat system** using Socket.IO with room-based event handling for concurrent user sessions.
- Integrated **Razorpay** and built a **digital wallet** with server-side order creation and payment signature verification.
- Implemented **authentication and authorization** middleware across REST APIs and optimized queries to reduce endpoint latency.

Projects

FileFlow
High-Performance File Transfer Engine | Tech Stack: React.js, Node.js, Socket.io, WebRTC Github

- Independently established a real-time file sharing platform using **Socket.io for low-latency signaling** (11 event listeners, 8 emitted events) and device discovery to establish **WebRTC peer-to-peer (P2P) data channels** for direct, zero-hop communication.
- Devised an **End-to-End (E2E) Ping-Pong ACK pipeline** to manage network backpressure, capping WebRTC `bufferedAmount` at **8MB** to guarantee zero-crash transfers for files up to **2GB**.
- Benchmarked throughput vs. overhead trade-offs to engineer a **4-tier dynamic chunk-sizing algorithm** (64KB → 1MB) that automatically scales binary payload sizes based on total file volume, drastically reducing CPU overhead and redundant network round-trips.
- Constructed a transaction-based engine with secure **session and room management**, ensuring data isolation, exact state synchronization, and integrity during concurrent multi-user file exchanges.
- Synchronized cross-device UIs using **event-driven progress tracking**, directly tying the sender's progress bar to the receiver's disk-write speed for accurate real-time latency representation.

WRDL – Real-Time Multiplayer Word Game
Next.js, React.js, Node.js, Express.js, Socket.IO, MongoDB, REST APIs, WebSockets, Tailwind CSS
Application: Github **Server:** Github

- Independently architected a **real-time multiplayer word-guessing game** supporting **2–4 players per room**, synchronizing game state across **15 Socket.IO event listeners** and **19 real-time broadcast events**
- Implemented **room-based architecture**, **live in-game chat**, and **turn-based gameplay** to handle concurrent multi-user interactions
- Engineered a **tiered scoring system** (20/15/10/5 pts by guess order, −10 for wrong guesses) alongside random word generation and hint logic to drive gameplay engagement
- Designed **3 MongoDB collections** (`User`, `Room`, `Word`) via Mongoose schemas and **5 REST API routes** for auth, room join/leave, and word retrieval

Extracurricular Activities

- Organized/co-organized 3+ coding contests and 2 technical events at IIIT Vadodara Coding Club, engaging 100+ participants